ANATOMY OF AI AGENTS 2026

www.arianprabowo.com · 10-Layer Agent Stack + Detached RAG/Multimodal Organism (Roitman arXiv:2606.24937)

What actually runs under the hood of an autonomous AI agent in 2026?
A 10-layer left-to-right spine: Silicon (L1) → uv/PyTorch/JAX (L2) → Data/Tokens (L3) → Arch/Pretrain (L4) →
SFT (L5) → RLHF/DPO/RLVR (L6) → Test-Time Compute (L7) → MCP (L8) → Harness^2 (L9) → Swarms (L10) + External RAG/MM.

① TRACE LEFT → RIGHT: Follow Core Layers 1 → 10 from silicon to swarms
② SEPARATE ORGANISM: Far-right column = detached RAG & Multimodal systems
③ HARDEN PROD: Audit the red FAILURE MODES & TRAPS row at bottom

*Synthesized from Haggai Roitman's 'The Hitchhiker's Guide to Agentic AI: From Foundations to Systems' (arXiv:2606.24937, v1.4, June 2026) + 2026 systems tooling. Companion to AI Skill Tree 2026 & AI Ladder Evolution by Arian Prabowo · Updated: 2026-11-07 23:57:39 AEDT (UTC+11:00)

AI AGENT STACK

LAYER 1 → LAYER 10

+ Detached Organism
(RAG & Multimodal)

PILLAR 1

CORE PRIMITIVES & ARCHITECTURE

Hardware Physics, Training Objectives, Inference Search & Agent Topologies

- L1-L3: Silicon/Env/Tokens
- L4-L7: Pretrain/SFT/RLVR/TTC
- L8-L10: MCP/Harness/Swarms

PILLAR 2

FRAMEWORKS, PKGS & PROTOCOLS

CLI Package Managers, DL/RL/Serving Engines, Agent Harnesses & Swarm Protocols

- brew · uv · conda · pip · venv
- PyTorch · JAX · TRL · verl · vLLM
- MCP · SKILL.md · LangGraph · A2A

PILLAR 3

SCALING, EVALS & SECURITY

Throughput SLAs, MicroVM Sandboxes, Verification Oracles & Zero-Trust Guardrails

- MFU · E2B/bwrap · Decontam
- RLVR Graders · PRM · MCP OAuth
- SWE-bench · pass^k · KYA

ANTI-PATTERNS

FAILURE MODES & TRAPS

What Breaks in Prod: Bottlenecks, Collapses, Reward Hacking & Swarm Cascades

- OOM & CUDA ABI Hell
- Catastrophic Forgetting & Hack
- Context Rot & O(N^2) Chatter

LAYER 1 →

PHYSICAL & SILICON

GPUs/TPUs/NPUs, HBM3e vs. SRAM hierarchy, CUDA/Triton kernels, and NVLink/InfiniBand fabrics.

- Core Unit: FLOPs, HBM Bytes & CUDA Warps
- Guide: Ch. 1.6, 2.1, 11.10-14, 30.5 (GPUs & Kernels).

GPU / TPU / NPU Tensor Cores
HBM3e vs. On-Chip SRAM Hierarchy
FlashAttention IO-Aware SRAM Tiling

- Streaming Multiprocessors (SMs)
- Arithmetic Intensity (FLOP/Byte)
- FP16 / BF16 / FP8 / FP4 Formats
- Unified Memory Architecture (UMA)
- Warp Schedulers (32-Thread SIMT)
- Roofline Compute vs. Memory Bound
- Online Softmax Rescaling (Safe Max)
- Wafer-Scale & SRAM-Only LPU Dies

- Register File (256KB/SM)
- HBM3e Bandwidth (3.3-8 TB/s)
- Stochastic Rounding
- Memory Coalescing
- L1 Cache / Shared Memory
- SRAM Bandwidth (~20-30 TB/s)
- Tensor Memory Accelerator (TMA)
- Bank Conflict Avoidance
- L2 Cache (50MB-128MB)
- Transformer Engine FP8/FP4
- Systolic Array Matrix Multiply
- TDP Power & Liquid Cooling

NVIDIA H100 / H200 / Blackwell B200
Google TPU v5p / v6e Trillium / v7
CUDA · OpenAI Triton · Apple Silicon M4

- NVIDIA GB200 NVL72 Rack Scale
- AWS Trainium2 & Inferentia2
- NVLink 4/5 (900-1,800 GB/s/GPU)
- cuBLAS / cuDNN / CUTLASS / FlashInfer
- AMD Instinct MI300X / MI325X
- Groq LPU & Cerebras WSE-3 ASIC
- InfiniBand NDR/XDR (400-800Gb/s)
- AMD ROCm / HIP & Apple Metal

- DGX / HGX 8-GPU Node Topology
- 80GB / 141GB / 192GB VRAM Tiers
- On-Device NPU (45-80+ TOPS)
- Liger-Kernel Fused Triton Ops
- GPUDirect RDMA / Storage
- Mac Mini M4 64GB-512GB Cluster
- Co-Packaged Silicon Photonics
- ECC Memory Error Correction
- Mellanox ConnectX-7 / 8 NICs
- RTX 4090 / 5090 Local Workstation
- Vera Rubin NVL144 (HBM4 Roadmap)
- NCCL / Gloo Collective Backend

Model FLOPs Utilization (MFU > 45%)
Prefill (Compute) vs. Decode (BW)
Confidential Compute GPU TEEs

- Prefill Phase (Matrix-Matrix Gemm)
- Multi-Instance GPU (MIG) Slices
- NVIDIA DCGM Hardware Health Check
- Nsight Compute & Systems Profiling
- Decode Phase (Matrix-Vector Gemv)
- Rail-Optimized Fat-Tree Topology
- Silent Data Corruption (SDC) Scans
- Hardware Root-of-Trust Attestation

- MFU = Achieved / Peak FLOPs
- 4-Model RLHF VRAM Multiplier
- NVLink All-Reduce Bandwidth
- H100/B200 Secure Enclave (CC-On)
- Hardware FLOPs Utilization (HFU)
- Optimizer State Memory (12B/param)
- Cross-Node All-to-All Bottleneck
- Power Usage Effectiveness (PUE)
- Bytes/Param: 2B (BF16) vs 0.5B (FP4)
- Activation Memory Checkpointing
- Thermal Throttling Mitigation
- Straggler GPU Detection

HBM Memory Bandwidth Wall (Decode)
CUDA Out-of-Memory (OOM) Crashes
Silent Tensor Core Bit-Flip Corrupt

▸ Low Decode MFU (<5% Compute Util)
▸ PCIe Host-to-Device Bottleneck
▸ InfiniBand Link Flaps & Stalls
▸ Straggler GPU Halting 1,000 Nodes

- 4-Model RLHF VRAM Explosion
- Unaligned Tensor Shapes (No TC)
- Thermal Throttling Under RL Rollouts
- Consumer GPU P2P Disabled Locks

LAYER 2 →

ENV, PACKAGES, PYTHON & TENSOR

Package managers (`Homebrew`, `uv`, `conda`, `pip`, `venv`), `Python`, `PyTorch`, `JAX`, `vLLM`, and sandboxes.

- Core Unit: Wheels, Lockfiles, Tensors & Paged KV
- Guide: Ch. 1.14-15, 11.1-9, 21.5, 22.1-9 (Env & Runtime).

Virtual Environments, Locks & Sandboxes
Autograd Computation Graphs & JIT
PagedAttention KV & Continuous Batch

- POSIX PTYs, Processes & Asyncio Loop
- OS Namespaces, cgroups & Seccomp
- PagedAttention Virtual Memory KV
- Disaggregated Prefill-Decode Serving
- ABI / manylinux Wheels & uv.lock
- Dynamic Autograd DAG & torch.compile
- RadixAttention Automatic Prefix Cache
- Constrained Grammar / FSM Masking

- pyproject.toml (PEP 517 / 621)
- LD_LIBRARY_PATH / DYLD RPATH
- XLA HLO Graph Fusion & CUDA Graphs
- Container UnionFS / OverlayFS Layers
- Content-Addressable Global Cache
- torch.compile (TorchInductor/Dynamo)
- Chunked Prefill Scheduling
- Hardware Virtualization (KVM / HVF)
- Hardlink / Reflink Env Cloning
- JAX jit / vmap / pmap / shard_map
- KV-Cache FP8 / INT4 Quantization
- IPC Shared Memory (/dev/shm)

Homebrew (brew) · uv · conda · pip · venv
Python 3.12+ · PyTorch 2.x · JAX · MLX
vLLM · SGLang · llama.cpp · Docker · E2B

- Homebrew (brew) & System Toolchains
- conda / Miniforge / mamba / Pixi / Poetry
- PyTorch 2.x · JAX / Flax · Apple MLX
- Docker / OCI & NVIDIA Container Toolkit
- uv (uv venv / uv run / uvx / uv pip)
- pip · venv · virtualenv · pipx
- vLLM · SGLang · TensorRT-LLM · Ollama
- Rust (PyO3/Maturin) & TypeScript/Bun

- brew install uv git ripgrep fd tmux
- pip install --no-build-isolation
- Outlines & XGrammar (JSON/CFG)
- Nix / Devcontainers / Direnv
- uv lock & uv sync --frozen
- DeepSpeed ZeRO & PyTorch FSDP2
- Ray Serve & NVIDIA Triton Server
- Slurm / Kubernetes (K8s) / Ray
- environment.yml & conda-lock
- llama.cpp (GGUF) & ExLlamaV2/v3
- ONNX Runtime & Safetensors Zero-Copy
- Pydantic v2 (Rust Schema Core)

E2B / Firecracker MicroVM Sandboxes
OS Sandboxing (Bubblewrap / Seatbelt)
TTFT / TPOT SLAs & Async RL Sync

- gVisor User-Space Kernel Isolation
- OpenEnv Standardized Gym Reset/Step
- Scoped Ephemeral OAuth / API Tokens
- Per-Tenant LoRA Adapter Hot-Swapping
- macOS sandbox-exec & Linux bwrap
- Air-Gapped VLAN & Egress Firewall
- NCCL Weight Broadcast (Trainer->vLLM)
- Grammar-Constrained 100% Valid JSON

- env.reset() / env.step(action) API
- TTFT < 200ms · TPOT < 15ms/tok
- S-LoRA / Punica Multi-Adapter Batch
- SBOM & pip-audit Vulnerability Scan
- SWE-bench Docker Repo Images
- Goodput (SLO-Compliant Tok/s)
- Pyodide / Wasm Code Execution
- Signed Container Attestation (Cosign)
- OSWorld / WebArena Live Desktop VMs
- In-Flight Weight Update (Zero Stall)
- Secret Scrubbing in Env Vars
- Deterministic Seed Mode (cuDNN)

Python Dependency & CUDA ABI Hell
Unsandboxed Host RCE & rm -rf /
KV-Cache Fragmentation & Preemption

▸ nvcc vs. torch.version.cuda Mismatch
▸ Global System Python Pollution
▸ pip install Breaking flash-attn ABI
▸ Typosquatted PyPI / npm / Skill Packs

- Agent Leaking .env & SSH Keys
- RL Rollout Generation Bottleneck
- 60-80% Contiguous KV Memory Waste
- torch.compile Graph-Break Thrashing

LAYER 3 →

DATA & TOKENIZATION

Pre-training/post-training corpora, BPE/Tiktoken tokenizers, deduplication, filtering, and synthetic data.

- Core Unit: Corpora, Subword Tokens & Shards
- Guide: Ch. 1.2, 3.2-3.5, 4.3, 6.2, 30.2 (Data & Tokenizers).

Web, Code & Math Corpus Curation
Byte-Level BPE & Unigram Tokenizers
MinHash Dedup & Quality Filtering

- Byte-Level BPE / SentencePiece / Tiktoken
- Vocabulary Size Scaling (32K-256K)
- Heuristic & FastText Quality Classifiers
- Synthetic Reasoning & Trajectory Gen
- Pre-Tokenization Regex Split Rules
- MinHash LSH & Suffix-Array Dedup
- Domain & Curriculum Mixture Weights
- Execution-Verified Code & Math Traces

- Tokenizer Fertility (Tokens/Word)
- Fill-in-the-Middle (FIM) Code Formatting
- Exact Substring 50-Token Dedup
- Compile/Test Sandbox Filtering
- Byte-Fallback for Unknown UTF-8
- AST-Parsed Repository Level Ordering
- Back-Translation & Persona Diversity
- Annealing Data Mix (High-Quality End)
- ChatML Special Control Tokens
- KenLM Perplexity Outlier Filtering
- Evol-Instruct Complexity Mutation
- Zero-Copy Memory-Mapped Shards

HF Datasets · Datatrove · NeMo Curator
Tiktoken · SentencePiece · HF Tokenizers
FineWeb-Edu · Dolma · The Stack v2

- HuggingFace Datasets & Hub Streaming
- Tiktoken & HuggingFace Rust Tokenizers
- The Stack v2 & StarCoderData (Code)
- Apache Arrow · Parquet · WebDataset
- Datatrove & NeMo Curator Pipelines
- FineWeb-Edu · Dolma · RedPajama-v2
- OpenMathReasoning & NuminaMath
- Distilabel & Curator Synthetic Suites

- Polars · DuckDB · PyArrow In-Memory ETL
- Megatron-LM IndexedDataset (.bin/.idx)
- CommonCrawl WARC / WET Extraction
- DVC & LakeFS Data Version Control
- Ray Data Distributed Streaming Ingest
- SWE-Gym & SWE-smith Issue Trajectories
- Trafilatura & Resiliparse HTML Clean
- Cleanlab Automated Label Auditing
- MosaicML StreamingDataset (MDS)
- OpenThoughts & Bespoke-Stratos Sets
- Tree-sitter AST Code Filtering
- Argilla Human-in-the-Loop Curation

Benchmark Decontamination (N-Gram)
PII, Secret & License Scrubbing
Synthetic Data Diversity & Entropy

- 13-Gram & Embedding Decontamination
- Regex + Presidio PII Scrubbing
- SPDX Open-Source License Verification
- Vendi Score & N-Gram Diversity Audits
- Canary String Injection & Filtering
- TruffleHog / Gitleaks Key Removal
- LLM-as-a-Judge Data Quality Scoring
- Tokenizer Compression Ratio Audits

- MMLU / GSM8K / HumanEval Overlap Scan
- Copyright & Verbatim Memorization Check
- Poisoned Backdoor Trigger Scanning
- Differential Privacy (DP) Scrubbing
- SWE-bench Repo Commit-Date Cutoff
- Multilingual Token Fairness Audit
- Synthetic Mode-Collapse Entropy Check
- Dataset Card & Croissant Metadata
- Opt-Out & robots.txt Compliance
- Toxic & CSAM Safety Classifier Filter
- Data Provenance & Lineage Ledgers
- Deterministic Shard Shuffle Seeding

Garbage-In Synthetic Model Collapse
Test-Set Contamination Mirage
Tokenizer Glitch & Fertility Bloat

▸ Recursive Synthetic Training Collapse
▸ Benchmark Leakage Inflating SOTA
▸ PII & API Keys Baked Into Weights
▸ Unbalanced Domain Mix Starving Code

- Tokenizer Glitch Tokens (SolidGoldMagikarp)
- Duplicate Boilerplate Overfitting
- 3-5x Token Bloat on Non-English/Code
- Broken UTF-8 / HTML Artifacts in Text

LAYER 4 →

ARCHITECTURE & PRE-TRAINING

Transformer/MoE/hybrid architectures (`GQA`, `MLA`, `Mamba/DeltaNet`), next-token pre-training, and scaling laws.

- Core Unit: Transformer Blocks, MoE & Base Weights
- Guide: Ch. 1, 2.2, 11.1-8, 30.1 (Architecture & Pre-Training).

Transformer · Sparse MoE · GQA / MLA
Hybrid Linear Attention (Mamba/DeltaNet)
Next-Token Cross-Entropy & Scaling Laws

- Pre-Norm RMSNorm & SwiGLU FFN
- Sparse MoE Top-K & Shared Experts
- RoPE & YaRN Context Extension (1M+)
- Chinchilla Compute-Optimal Scaling
- Grouped-Query (GQA) & Latent (MLA)
- Gated DeltaNet / Mamba-2 SSM Hybrid
- Autoregressive Next-Token Objective
- 3D Parallelism (DP / TP / PP / SP / EP)

- Multi-Head Latent Attention (KV Compress)
- Sliding Window & Ring Attention
- ZeRO Stage 1/2/3 & FSDP2 Sharding
- muP (Maximal Update Parametrization)
- Aux-Loss-Free MoE Load Balancing
- AdamW vs. Muon Matrix Orthogonal
- 1F1B & Interleaved Pipeline Schedule
- QK-Norm Attention Logit Stabilization
- Multi-Token Prediction (MTP) Heads
- WSD (Warmup-Stable-Decay) LR Schedule
- FP8 Mixed-Precision Pre-Training
- BitNet b1.58 Native Ternary Architecture

Megatron-LM · DeepSpeed · TorchTitan
Llama 4 · Qwen 3 · DeepSeek-V3 Base
HuggingFace Transformers · OLMo 2

- NVIDIA Megatron-Core & NeMo Framework
- DeepSpeed ZeRO-3 & Megatron-DeepSpeed
- Nanotron & LitGPT Pre-Training Suites
- Qwen3 & Llama-4 Pre-Trained Base Models
- PyTorch TorchTitan & FSDP2 Native
- MosaicML Composer & LLM-Foundry
- DeepSeek-V3 (671B/37B MLA+MoE Base)
- OLMo-2 / Marin Fully-Open Base Checkpoints

- GPT-5.4 / Claude 4.6 / Gemini 3 Base
- Nemotron-H Hybrid Mamba-Transformer
- TransformerEngine FP8 Linear Layers
- AWS ParallelCluster & Slurm Orchestrator
- Mistral Large 3 & Codestral Base
- FlashAttention-3 Hopper FP8 Kernels
- HuggingFace Accelerate & Safetensors
- Checkpointer Async S3 / Lustre Write
- Qwen3-Next Hybrid Gated-DeltaNet
- DeepEP High-Throughput MoE All-to-All
- WandB · TensorBoard · MLflow Tracking
- EleutherAI GPT-NeoX Legacy Reference

Loss-Spike Recovery & Gradient Clipping
Base Perplexity (Bits-per-Byte) & NIAH
Distributed Checkpoint & Elastic Faults

- Global Gradient Norm Clipping (1.0)
- Automated Loss-Spike Batch Skip/Rewind
- Needle-in-a-Haystack (NIAH) & RULER
- Async Distributed Checkpointing (<5s)
- Z-Loss Router Logit Regularization
- Validation Bits-per-Byte (BPB) Tracking
- Few-Shot Base Evals (MMLU / HumanEval)
- muP Proxy-Model Hyperparameter Transfer

- Activation Outlier & Kurtosis Monitoring
- FP8 Underflow / Overflow Scaling Audit
- Continued Pre-Training (CPT) Replay Mix
- Checkpoint Integrity SHA-256 Signing
- MoE Expert Utilization Entropy Audit
- In-Flight Straggler & NaN Detector
- Base Model Memorization Extraction Audit
- Scaling Law Extrapolation (Loss vs FLOPs)
- Dead-Neuron & Attention-Sink Diagnostics
- Cross-Datacenter DiLoCo Low-Comm Sync
- CBRN / Dual-Use Pre-Training Data Filter
- Long-Context Perplexity Decay Curve

Catastrophic Loss Spikes & NaN Divergence
MoE Routing Collapse (Expert Starvation)
Pipeline Bubble & All-to-All Bottlenecks

▸ BF16/FP8 Attention Logit Overflow
▸ Top-K Router Dumping All Tokens to 1 Expert
▸ 30-50% GPU Idle Time in Naive Pipeline
▸ Cross-Node EP All-to-All Network Saturation

- Attention Sinks Hoarding Mass on BOS
- Catastrophic Forgetting in Domain CPT
- RoPE Out-of-Distribution Length Collapse
- Corrupted Checkpoint From Silent SDC

LAYER 5 →

SUPERVISED FINE-TUNING (SFT)

Instruction tuning, ChatML formatting, sequence packing, completion-only loss masking, and `PEFT` (`LoRA`/`DoRA`).

- Core Unit: Chat Templates, Packed Sequences & LoRA
- Guide: Ch. 1.7-9, 3, 4, 30.2 (SFT, Packing & PEFT Adapters).

Instruction Tuning & ChatML Templates
Sequence Packing & Completion Masking
PEFT Adapters (LoRA / QLoRA / DoRA)

- ChatML Roles (System / User / Assistant / Tool)
- Multipack / FlashAttention Varlen Packing
- LoRA (B*A Low-Rank Adapter Matrices)
- DoRA (Weight-Decomposed Magnitude/Dir)
- Completion-Only Loss Masking (Labels=-100)
- Position-ID Reset Across Packed Examples
- QLoRA (4-Bit NF4 Base + Double Quant)
- PiSSA & rsLoRA (Rank-Stabilized alpha/sqrt(r))

- Target Modules: q/k/v/o/gate/up/down_proj
- Zeroth-Order MeZO Forward-Only Tuning
- Rejection Sampling SFT (Best-of-N Filter)
- Model Merging (SLERP / TIES / DARE)
- LoRA Rank r = 8..128 & Alpha Scaling
- GaLore Gradient Low-Rank Projection
- Multi-Turn Tool-Call Trajectory SFT
- Adapter Fusion & Task Arithmetic
- NEFTune Noisy Embedding Regularization
- LISA Layerwise Importance Sampled Adam
- Harness Distillation (Teacher->Student SFT)
- Spectrum SNR-Guided Selective Layer Freeze

HF TRL (SFTTrainer) · PEFT · Unsloth
Axolotl · LLaMA-Factory · Torchtune
MergeKit · bitsandbytes · Liger-Kernel

- HuggingFace TRL SFTTrainer & DataCollator
- Unsloth (2x Speed, 70% Less VRAM Triton)
- PyTorch Torchtune & MosaicML Finetuning
- Liger-Kernel Fused Cross-Entropy Loss
- HuggingFace PEFT (LoraConfig / get_peft_model)
- Axolotl & LLaMA-Factory YAML SFT Suites
- bitsandbytes NF4 & HQQ / AWQ Quantizers
- MergeKit CLI Multi-Adapter Checkpoint Merger

- DataCollatorForCompletionOnlyLM
- LoRA+ (Asymmetric A/B Learning Rates)
- OpenAI Harmony & ShareGPT JSONL Schemas
- Predibase Lorax Multi-Tenant Server
- packing=True + flash_attention_2 Varlen
- VeRA (Vector-Based Random Matrix Adaptation)
- S-LoRA & Punica Multi-Adapter Serving
- HuggingFace Alignment Handbook Recipes
- apply_chat_template(tokenize=True)
- AdaLoRA SVD Adaptive Rank Allocation
- vLLM --enable-lora Dynamic Adapter Load
- OpenPipe / Fireworks SFT Cloud Pipelines

Masking Verification (Zero Prompt Loss)
IFEval Instruction-Following & Format Audit
Safety Refusal & Backdoor Trigger Scans

- Prompt-Token Loss Masking Assertion (-100)
- Cross-Sample Attention Bleed Check (Varlen)
- MT-Bench & AlpacaEval 2.0 Win-Rate Audits
- Benign vs. Harmful Refusal Calibration Mix
- EOS / Stop-Token Generation Verification
- IFEval Strict Format & Constraint Benchmarks
- Base-Capability Retention (MMLU / Code Drop)
- Fine-Tuning Jailbreak & Sleeper-Agent Audit

- Gradient Norm & Eval-Loss Early Stopping
- Tool-Call JSON Schema Validity Rate
- Shadow-Alignment Superficial Style Check
- PII Memorization Canary Extraction Test
- Cosine LR with 3-10% Warmup Ratio
- merged_and_unload() FP16 Export Check
- Emergent Misalignment Safety Probe
- Multi-Turn Context-Switch Coherence Eval
- Epoch Cap (1-3 Epochs to Prevent Overfit)
- Per-Token Log-Probability Drift Check
- LoRA Adapter Checksum & Provenance
- Quantized QLoRA vs. BF16 LoRA Parity Check

Training on Prompt Tokens (Missing -100)
Chat Template EOS/BOS Mismatch Loops
Catastrophic Forgetting & Style Overfit

▸ Unmasked User Prompts Corrupting Policy
▸ Wrong EOS Token Causing Infinite Babbling
▸ Naive Packing Bleeding Attention Across Docs
▸ Applying LoRA Only to Q/V (Missing MLP)

- Overfitting Small SFT Set (>3 Epochs)
- 100 Safety Examples Breaking Benign Help
- Superficial Alignment (Style Without Logic)
- Narrow Insecure-Code SFT Misaligning Model

LAYER 6 →

POST-TRAINING (RLHF, DPO & RLVR)

Preference alignment (`PPO`, `DPO`, `KTO`) and critic-free verifiable-reward RL (`RLVR`: `GRPO`, `Dr. GRPO`, `DAPO`).

- Core Unit: Preference Pairs, Rollouts & Advantages
- Guide: Ch. 5, 6, 7, 8, 9, 10, 12, 13 (RLHF, DPO & RLVR).

Preference Tuning: PPO · DPO · KTO · SimPO
Critic-Free RLVR: GRPO · Dr. GRPO · DAPO
Verifiable Execution Oracles & PRMs

- Bradley-Terry Reward Model & PPO (GAE)
- KTO (Prospect Theory Unpaired Binary Feedback)
- GRPO (Group-Relative Advantage, No Critic)
- DAPO (Decoupled Clip-Higher & Dynamic Sampling)
- Direct Preference Optimization (DPO Implicit RM)
- IPO · ORPO · SimPO (Reference-Free Length Norm)
- Dr. GRPO (Unbiased Length & Std Normalization)
- Cold-Start Pure-RL Emergence (DeepSeek-R1-Zero)

- KL Penalty beta * D_KL(pi_theta || pi_ref)
- Outcome RM (ORM) vs. Process RM (PRM)
- REINFORCE++ & RLOO Leave-One-Out Baseline
- Rule-Based Format + Execution Reward Sum
- PPO Clipped Surrogate epsilon in [0.1, 0.28]
- RLCR (Confidence Calibration Brier Reward)
- Iterative Online DPO & Spin Self-Play
- Curriculum Difficulty Filtering (0 < Pass < 1)
- GRPO Group Size G = 8..64 Rollouts/Prompt
- On-Policy Self-Distillation (OPSD / RLCSD)
- Constitutional AI (RLAIF Critique-Revise)
- Multi-Turn Agentic RL Environment Rollouts

verl · OpenRLHF · HF TRL (GRPO / DPO)
DeepSeek-R1 · OpenAI o3/o4 · QwQ-32B
Prime Intellect verifiers · NeMo Aligner

- verl (Volcengine HybridEngine RLHF/RLVR)
- HuggingFace TRL (GRPOTrainer / DPOTrainer)
- Unsloth Fast GRPO (Single-GPU RLVR)
- AllenAI Open-Instruct (TÜLU 3 RLVR Pipeline)
- OpenRLHF (Ray + vLLM Distributed Async RL)
- Prime Intellect verifiers & OpenR1 Suites
- NVIDIA NeMo Aligner & Megatron-LM RL
- SkyRL &oat-rl Distributed Agentic RL

- vLLM Rollout Engine + FSDP2 Actor Sync
- Pytest / Sandbox Code Execution Graders
- RewardBench & RM-Bench Reference Models
- Ray PlacementGroups for Actor/Ref/Rollout
- NCCL In-Flight Weight Broadcast
- SWE-RL & WebRL Agent Trajectory Trainers
- ArmoRM Multi-Objective Reward Model
- WandB Rollout Length & Entropy Dashboards
- Lean 4 / SymPy Formal Math Verifiers
- Search-R1 Multi-Turn Retrieval RL
- Llama-3.1-Nemotron-70B-Reward
- DeepSeek-R1-Distill (1.5B-70B Checkpoints)

Deterministic Sandboxed Reward Graders
KL-Divergence & Policy Entropy Monitoring
Anti-Reward-Hacking & Alignment Audits

- Hermetic Unit-Test & Math Answer Verifiers
- Policy Entropy & KL-Divergence Guardrails
- RewardBench & JudgeBench Grader Audits
- Deliberative Alignment in Reasoning Traces
- DAPO Dynamic Filtering of All-0 / All-1 Groups
- Length-Controlled Win-Rate (AlpacaEval LC)
- Instruction Hierarchy Privilege Training
- Sandbagging, Sycophancy & Deception Audits

- Rollout Generation Wall-Clock (70-90% of RL)
- Code Grader Timeout & Anti-Tamper Lock
- Activation Steering & SAE Circuit Probes
- CBRN / Cyber ASL-3/4 Capability Gates
- Off-Policy Staleness Bound in Async RL
- Over-Optimization Goodhart Curve Detection
- Machine Unlearning (RMU / Gradient Ascent)
- Expected Calibration Error (ECE) Post-RL
- Format Reward Weight <= 0.1 vs. Accuracy 1.0
- Pass@1 vs. Pass@64 Diversity Gap Check
- GCG / AutoDAN / Best-of-N Jailbreak Evals
- Contamination-Free LiveBench & AIME Evals

Reward Hacking & Length Inflation
Policy Entropy Collapse & Mode Collapse
GRPO Zero-Variance Advantage (All 0/1)

▸ Model Gaming Regex/Judge Without Solving
▸ GRPO Group All-Correct/All-Wrong -> 0 Grad
▸ DPO Likelihood Displacement (Chosen Drops)
▸ Std-Dividing Advantage Biasing Easy/Short

- Policy Entropy Crashing to 0 (Repetitive CoT)
- Stale vLLM Rollout Weights Breaking PPO Ratio
- 4-Model PPO OOM Without Hybrid Sharding
- Language Mixing in Unpenalized CoT Traces

LAYER 7 →

TEST-TIME COMPUTE & REASONING

Inference-time System-2 scaling: Chain-of-Thought (`CoT`), thinking budgets, `Best-of-N`, `PRM` reranking, and `MCTS`.

- Core Unit: Thinking Tokens, Search Trees & Votes
- Guide: Ch. 12.1-8, 13, 19.2, 30.3 (Test-Time Compute & Search).

System-2 Chain-of-Thought (<think> CoT)
Parallel Scaling: Best-of-N & Majority@k
Sequential & Tree Search (PRM / MCTS)

- Long Internal Reasoning (<think>...</think>)
- Self-Consistency Majority Voting (maj@k)
- Step-Level Beam Search Guided by PRM
- Sequential Self-Correction & Reflexion Loops
- Adaptive Thinking Token Budgets (Low/Med/High)
- Best-of-N Reranking With Outcome/Process RM
- Monte Carlo Tree Search (MCTS / UCT Value)
- Compute-Optimal Test-Time Scaling Allocation

- PUCT = Q(s,a) + c * P(s,a) * sqrt(N)/(1+n)
- Budget Forcing ('Wait' vs. '</think>' Token)
- Self-Verification & Backtracking Traces
- Confidence-Informed Early Stopping
- Tree-of-Thoughts (ToT) BFS/DFS Branching
- Entropy-Guided Dynamic Branching
- Pass@k Upper-Bound vs. Majority@k Gap
- Iterative Code-Execution Refine Loops
- Graph-of-Thoughts (GoT) Aggregation
- Speculative Reasoning (Big Draft / Small Run)
- Difficulty-Aware Compute Routing
- Test-Time Training (TTT) Local Weight Update

OpenAI o3 / GPT-5.4 ReasoningEffort API
Claude 4.6 Extended Thinking · Gemini 3
DeepSeek-R1 · Qwen3 Thinking · s1 Budget

- OpenAI reasoning_effort (low / medium / high)
- Gemini 2.5/3 Thinking Budget & Thought Summary
- vLLM / SGLang Reasoning Parser (--reasoning)
- Search-and-Learn (HuggingFace SAL Recipes)
- Anthropic Extended Thinking (budget_tokens)
- Qwen3 Hybrid Thinking (/think vs. /no_think)
- Optillm Inference-Time Scaling Proxy (MCTS/BoN)
- LATS (Language Agent Tree Search) Engine

- Qwen2.5-Math-PRM-7B / Shepherd PRM
- DeepConf Entropy-Filtered Voting
- vLLM Prefix Caching Across Parallel Samples
- DSPy BestOfN & Refine Inference Modules
- Skywork-o1-Open-PRM Step Verifiers
- AlphaProof / AlphaCode 2 Search Pipelines
- SGLang Fork/Join Parallel Branching DSL
- LiteLLM Unified Thinking-Token Billing
- s1 Simple Test-Time Scaling (Budget Forcing)
- Archon Compound Inference-Time Architecture
- LeanDojo / Pantograph Proof-Tree Search
- OpenRouter Reasoning-Token Stream API

Accuracy-per-Log-FLOP Scaling Curves
Pass@k vs. Cons@k & PRM Calibration
Overthinking & CoT Faithfulness Audits

- AIME / HMMT / Codeforces Rating vs. Token Budget
- Pass@k vs. Majority@k / PRM@k Efficiency
- Token-Budget Cap & Diminishing-Return Cutoff
- Hidden vs. Visible Reasoning Trace Safety
- ARC-AGI-1/2 & Humanity's Last Exam (HLE)
- ProcessBench Step-Error Localization Audit
- CoT Faithfulness & Post-Hoc Rationalization
- Latency-SLA Routing (Fast System-1 vs. System-2)

- Optimal Compute Allocation (Smallx64 vs Bigx1)
- Self-Correction Positive vs. Negative Flip Rate
- Monitorability of Unfaithful CoT Scheming
- Deterministic Voting Tie-Breaking Rules
- First-Correct-Solution Token Index
- PRM Step-Level AUROC & Calibration
- Reasoning-Token Cost Circuit Breakers
- Verifier False-Positive Rate on Hard Tail
- Thinking-to-Answer Token Ratio Monitoring
- Prefix-Cache Hit Rate During Tree Search
- Prompt-Injected Infinite-Thinking DoS Guard
- LiveCodeBench Holistic Contamination-Free

Overthinking & Analysis Paralysis (10x Cost)
PRM Reward Hacking & Verifier Blindspots
Self-Correction Degrading Correct Answers

▸ Burning 32K Thinking Tokens on Trivial Queries
▸ Circular 'Wait, let me re-verify' Babble Loops
▸ Majority Vote Converging on Shared Bias
▸ Imperfect PRM Selecting Adversarial Fluff

- Intrinsic Self-Correction Without Oracle Hurting
- Tree Search Exploding KV-Cache VRAM (O(B^D))
- Unfaithful CoT Hiding True Decision Shortcut
- Reasoning-Effort Timeout Dropping Tool Calls

LAYER 8 →

CONTEXT, MEMORY, TOOLS & MCP

Context budgeting, 4-tier agentic memory (`Sleep-Time Compute`), function calling, `MCP`, and `SKILL.md`.

- Core Unit: Context Windows, Memory & JSON-RPC
- Guide: Ch. 17, 18.2-3, 23, 24 (Memory, Tools, MCP & Skills).

Context Engineering & Token Budgeting
4-Tier Memory + Sleep-Time Dreaming
Tool Calling, MCP Bus & Progressive Skills

- Working Memory (Active Context Window)
- Semantic Memory (Entity/Fact Triples)
- Sleep-Time Offline Consolidation (Dream)
- Progressive Disclosure Skill Loading
- Episodic Memory (Past Experience Logs)
- Procedural Memory (SKILL.md / Weights)
- JSON-RPC 2.0 Host-Client-Server Bus
- Observation Masking & /compact Hygiene

- System / History / Scratchpad Token Budget
- Proactive Context Pre-Fetching
- Strict JSON Schema Tool Definitions
- Rolling Summary & Turn Truncation
- Write / Read / Reflect / Consolidate Ops
- KV-Cache Prefix Pinning (90% Discount)
- Code-Act (Python Execution as Tool)
- Subagent Forked Context Isolation
- Ebbinghaus Temporal Memory Decay
- Parallel Tool-Call Batch Execution
- Dynamic Tool Search (Top-K Schemas)
- Scratchpad & Todo.md State Offload

Model Context Protocol (MCP / .mcp.json)
Agent Skills (SKILL.md Folder Standard)
Letta (MemGPT) · Mem0 · Zep · LangMem

- MCP Primitives: Tools, Resources, Prompts
- FastMCP Python / TypeScript SDKs
- skills/scripts/ & skills/references/ Dirs
- Mem0 · Zep · Graphiti Temporal Memory
- MCP Transports: stdio & Streamable HTTP
- SKILL.md YAML Frontmatter + Instructions
- Letta (MemGPT) OS-Paged Memory Blocks
- Linux Foundation AAIF MCP Registry

- MCP Sampling & Roots Client Callbacks
- agentskills.io Open Specification
- Redis / Postgres Agent Session Stores
- GitHub / Slack / Postgres MCP Servers
- MCP Elicitation User Confirmation
- LangMem Semantic/Episodic Store
- OpenAPI-to-MCP Auto-Generators
- Anthropic & Cursor MCP Marketplaces
- .claude/skills/ & .agents/skills/ Paths
- Cognee & Memary Memory Engines
- Browserbase / Playwright MCP Servers
- Smithery & Glama MCP Registries

MCP OAuth 2.1 Scopes & Tool Signing
/compact Hygiene (<65% Context Cap)
Memory Contradiction & Decay Audits

- Human-in-the-Loop Destructive Tool Gate
- Memory Contradiction & Stale-Fact Purge
- MCP Server Code-Signing Verification
- OX Security stdio RCE Hardening
- Tool Schema Token-Bloat Minimization
- LOCOMO & LongMemEval Memory Evals
- Least-Privilege Read-Only Connectors
- Cross-Tenant Memory Isolation

- Context Utilization < 65% Rot Threshold
- ToolBench & API-Bank Multi-Tool Eval
- Skill Static Analysis Before Install
- Idempotency Keys for Mutating Tools
- Lost-in-the-Middle U-Curve Placement
- Episodic Replay Buffer Deduplication
- Egress Domain Whitelisting per Tool
- Audit Logging of All JSON-RPC Calls
- BFCL (Berkeley Function-Calling) Eval
- Tool Output Truncation & Pagination
- OAuth 2.1 PKCE Token Refresh Flow
- Canary Tokens in Memory Stores

Context Rot & Lost-in-the-Middle
MCP Tool Bloat (100+ Schemas)
Malicious Skills & MCP Rug-Pulls

▸ 100+ MCP Tool Schemas Blowing Context
▸ ClawHavoc Malicious SKILL.md Payloads
▸ MCP Rug-Pull & Tool Shadowing Attacks
▸ Memory Poisoning Across Sessions

- Stale Episodic Memory Overriding Code
- Malformed Tool-Call JSON Loops
- stdio MCP Server Spawning Host Shells
- Unbounded Tool Output Flooding Window

LAYER 9 →

AGENT HARNESS, LOOP & GRAPH

The CPU/OS of an agent: `CLAUDE.md`/`AGENTS.md`, Loop Engineering, Graph Engineering (`LangGraph`), and `Harness²`.

- Core Unit: State Transitions, Actions & Diffs
- Guide: Ch. 18, 19, 20, 21, 27 (Harness, Loops & Graphs).

Agent Harness (Prompt+Tools+State+Loop)
Loop Engineering (Observe→Decide→Act)
Graph Engineering (Typed StateGraphs)

- ReAct · Reflexion · Plan-and-Execute
- Typed State Schema & Reducer Functions
- Fan-Out / Fan-In Parallel Send() API
- Spec-Driven Development (Spec→Plan→Task)
- 5 Loop Primitives (State/Step/Check/Stop)
- Conditional Edges & Cyclic Graph Routing
- Recursive Agent Harnessing (Harness^2)
- Cognitive Escalation & Uncertainty Handoff

- pi(a_t | s_t) Inference-Time Policy Loop
- BDI (Belief-Desire-Intention) Hybrid
- Stagnation & Repeat-Action Detection
- Exponential Backoff & Schema Self-Repair
- LATS (Language Agent Tree Search)
- Checkpointer Time-Travel & Replay
- Subagent Context Isolation (Forked Call)
- Token/Cost/Iteration Termination Caps
- LLM Compiler Parallel DAG Execution
- Human-in-the-Loop Interrupt() Gates
- Red-Green-Refactor TDD Agent Cycle
- Dynamic Model Routing (Fast vs. Reasoner)

Claude Code CLI · Codex CLI · OpenCode
Cursor · Windsurf · Cline / Roo · Aider
LangGraph · OpenAI Agents SDK · DSPy

- CLAUDE.md & AGENTS.md Repo Constitutions
- GitHub Spec Kit & obra/superpowers TDD
- Smolagents (CodeAct Python-as-Action)
- OpenHands (OpenDevin) & Block goose
- .cursor/rules & .github/copilot-instructions
- LlamaIndex Workflows (Event-Driven)
- PydanticAI Type-Safe Agent Harness
- Semantic Kernel & Haystack Agents

- LangGraph PostgresCheckpointer
- Browser-Use & Playwright DOM Harnesses
- Instructor & BAML Structured Outputs
- Repomix / Gitingest Codebase Packers
- DSPy MIPROv2 Prompt/Weight Compiler
- Vercel AI SDK Core (generateText/maxSteps)
- Litellm Unified Model Router
- LangSmith · Arize Phoenix · Langfuse
- Claude Agent SDK & Computer Use API
- Guardrails AI & NeMo Guardrails
- Git Pre-Commit & CI Harness Hooks
- OpenTelemetry (OTel) GenAI Spans

Decoupled Verification Oracles (No Self-Grade)
SWE-bench Verified/Pro & Terminal-Bench
pass^k Reliability & Trajectory Tracing

- Property-Based Testing (Hypothesis/Kani)
- Immutable CI Tests Locked From Agent Write
- LLM-as-a-Judge Pairwise Rubric Grading
- PyRIT & Garak Adversarial Red-Teaming
- Mutation Testing (Stryker / mutmut) Gates
- Trajectory Tool-Call Exact/Prefix Match
- Cost-per-Resolved-Task Efficiency Metric
- LlamaFirewall Prompt/Tool Guardrails

- pass^k = (pass@1)^k Consistency Metric
- Scale SWE-bench Pro (Private Repos)
- Shadow-Mode Traffic Counterfactual Eval
- Git Diff AST Security Linter (Semgrep)
- METR 50%-Reliability Horizon (2h-30h)
- Step Efficiency (Optimal / Actual Steps)
- Circuit-Breaker Spend & Loop-Depth Caps
- Automated Rollback on Regression
- GAIA · OSWorld · WebArena · TAU-bench
- Deterministic Mock-Tool Replay Harness
- Canary Token Exfiltration Tripwires
- EU AI Act / SOC2 Immutable Audit Logs

Agent Deleting / Mocking Failing Tests
Unbounded Loop Burn & Action Stagnation
God-Node Anti-Pattern & State Bloat

▸ Coupled Agent Rewriting Its Own Grader
▸ 0.95^20 = 35.8% Compounding Step Decay
▸ Poisoned .cursor/rules or GitHub Issues
▸ CVE-2025-53773 IDE Config Auto-Approve RCE

- Swallowing Exceptions in Tool Wrappers
- Premature Termination Before Verification
- Appending Raw 50KB HTML to Graph State
- Sycophantic Plan Pivot on User Pushback

LAYER 10 ★

MULTI-AGENT SWARMS & A2A

Multi-agent topologies (`OWC`), inter-agent protocols (`A2A`/`ACP`), parallel worktree swarms, and Agentic/Generative UI.

- Core Unit: A2A Tasks, Worktrees & GenUI Streams
- Guide: Ch. 21.6, 25, 26, 28, 30.10/18 (A2A, MAS & Agentic UI).

Orchestrator-Worker-Critic (OWC) Pattern
Multi-Agent Topologies (Hierarchical/Mesh)
Generative UI & Shared State Sync

- Centralized Supervisor vs. Decentralized
- Debate / Society-of-Mind Consensus
- Contract-Net Task Bidding & Auctions
- A2A (Horizontal) × MCP (Vertical) Dual Bus
- Hierarchical Tree (Squads of Squads)
- Blackboard Shared-Memory Coordination
- Parallel Git-Worktree Swarm Isolation
- Human-on-the-Loop Escalation & Approval

- Agent Card (/.well-known/agent.json)
- Credit Assignment (Shapley / Counterfactual)
- Pub/Sub Asynchronous Event Choreography
- Optimistic UI Updates & Diff Previews
- Task Lifecycle (Submitted→Working→Done)
- Role Specialization & Persona Diversity
- Process-Centric Enterprise State Workflows
- Ambient OS Cross-App Delegation
- Multi-Agent MDP & CTDE MARL Training
- Byzantine Fault-Tolerant Critic Voting
- Server-Driven UI Component Streaming
- Organizational Digital-Twin Simulation

Google A2A Protocol · IBM ACP · ANP
CrewAI · AutoGen / AG2 · LangGraph Swarm
CopilotKit · AG-UI · Vercel AI SDK UI

- Google Agent-to-Agent (A2A) JSON-RPC/SSE
- Microsoft AutoGen v0.4 / Semantic Kernel
- Claude Code Subagents & Worktree Swarms
- Agent Gateways (Kong / Solo / LiteLLM)
- IBM BeeAI & Agent Communication Protocol
- CrewAI Flows & Role-Based Crews
- OpenClaw 24/7 Multi-Channel Agent Daemon
- Claude Artifacts · ChatGPT/Gemini Canvas

- AG-UI (Agent-User Interaction Protocol)
- OpenAI ChatKit & Realtime WebRTC Voice
- OpenAI Swarm / Agents SDK Handoffs
- Slack / Discord / WhatsApp Bot Bridges
- Vercel AI SDK useChat / streamUI Hooks
- LangGraph Platform Cloud & Studio UI
- A2A Artifact & Push-Notification Webhooks
- Linear / GitHub PR Async Agent Teammates
- Chainlit · Streamlit · Gradio Agent UIs
- Camel-AI Workforce & MetaGPT SOPs
- React Server Components (RSC) GenUI
- 1-Click MCP & A2A Enterprise Registries

Agent Gateway (mTLS, Auth, FinOps Caps)
Cross-Agent Trust & Cryptographic KYA
Independent Critic & Red-vs-Blue Swarms

- Zero-Trust Agent Identity (SPIFFE / KYA)
- Per-Swarm Token & Dollar Circuit Breakers
- Cross-Repo API Contract Locks (OpenAPI)
- Provenance Watermarking (C2PA / SynthID)
- OAuth 2.1 On-Behalf-Of (OBO) Token Flow
- Anti-Collusion & Decorrelated Critic Models
- Deadlock, Livelock & Cycle Detection
- Strict Liability & Action Audit Ledgers

- Communication Overhead O(N^2) Pruning
- Swarm Amplification Factor Monitoring
- XBOW / Cybench / T3MP3ST Cyber Ranges
- Anti-Sycophancy & Anti-Psychosis Filters
- Multi-Agent Judge Independence Check
- Canary Deployment of Agent Topologies
- Lean 4 / TLA+ Formal Protocol Verification
- GDPR / HIPAA Cross-Agent Consent Scopes
- Cross-Org Data-Loss Prevention (DLP)
- Chaos Engineering (Injecting Rogue Agents)
- Human Escalation SLA Latency Tracking
- Kill-Switch Broadcast Across Swarm Nodes

Cascading Hallucination & Swarm Echo
O(N^2) Chatty Agent Token Explosion
Cross-Agent Confused Deputy & Escalation

▸ 7-Persona Roleplay Chatter Burning 15x Cost
▸ Homogeneous Critic Blindly Agreeing
▸ Supervisor Bottleneck & Context Overflow
▸ Merge Conflicts Across Parallel Worktrees

- Rogue Subagent Bypassing Parent Scopes
- Free-Riding Worker Agents in Crews
- Infinite Ping-Pong Handoff Deadlocks
- UI Stream Desync & Phantom Tool States

⇄ SEPARATE ORGANISM

EXTERNAL

RAG & MULTIMODAL (EXTERNAL)

Detached retrieval (`Hybrid/Graph RAG`) and sensory (`Vision/Audio/Omni`) subsystems that plug into the agent stack.

- Core Unit: Vector Chunks, ViT Patches & Audio Frames
- Guide: Ch. 14, 15, 16, 18.4, 30.8/9 (RAG, Vision & Audio).

Hybrid & Agentic RAG (Dense+BM25+Rerank)
GraphRAG Entity Summaries & REFRAG
Vision (ViT/SigLIP) & Native Omni Audio

- Bi-Encoder (HNSW) vs. Cross-Encoder Rerank
- Contextual Chunking & AST Code Splitting
- ViT Patch Projection & AnyRes Dynamic Tiling
- Whisper Spectrogram & RVQ Audio Codecs
- Reciprocal Rank Fusion (RRF) & ColBERT
- Self-RAG / Corrective RAG / Search-R1
- Visual Token Compression (Perceiver / Q-Former)
- GUI Coordinate Grounding (Set-of-Mark / SoM)

- HNSW / IVF-PQ / DiskANN Vector Indices
- GraphRAG Community Detection (Leiden)
- Chameleon / Gemini Early-Fusion Omni
- Full-Duplex Speech-to-Speech Streaming
- Matryoshka Representation Learning (MRL)
- REFRAG Compress-Sense-Expand Decoding
- 2D-RoPE Spatial Vision Positional Encoding
- Video Temporal Frame Sampling & 3D Tubes
- HyDE Hypothetical Document Embeddings
- SigLIP / CLIP Contrastive Alignment
- ColPali Visual Document Patch Retrieval
- Diffusion / Flow-Matching Visual Generation

Qdrant · Milvus · pgvector · Neo4j
LlamaIndex · Haystack · LightRAG · Docling
GPT-5.4 Omni · Gemini 3 · Qwen2.5-VL

- Qdrant · Milvus · Pinecone · Weaviate
- LlamaIndex · Haystack · Microsoft GraphRAG
- Docling · Unstructured · Marker · MinerU PDF
- Whisper-v3 · Moshi · ElevenLabs · Cartesia
- pgvector · LanceDB · Chroma · FAISS
- BGE-M3 · Cohere Embed v3 & Rerank v3.5
- Qwen2.5-VL / Qwen-Omni · Pixtral · InternVL
- Claude Computer Use & OpenAI Operator GUI

- Elasticsearch / OpenSearch BM25 Lexical
- Neo4j & FalkorDB Property Graph Stores
- OmniParser UI Icon/Bounding-Box Detector
- SAM 2 (Segment Anything) Spatial Masks
- Firecrawl & Jina Reader Markdown Scrapers
- SQLite-vec & DuckDB-VSS Local Vector
- Playwright + Screenshot Visual DOM Hybrid
- Flux.1 / SD3.5 / Veo 3 Media Generators
- ColPali / Byaldi Multimodal PDF RAG
- LiveKit & Pipecat Real-Time Voice/Video
- PaddleOCR & Surya Multilingual OCR
- TwelveLabs & LanguageBind Video Indexers

RAG Triad Evals (Faithfulness / Recall@k)
Document-Level RBAC & Citation Locking
MMMU / ScreenSpot / OSWorld GUI Evals

- Ragas & TruLens Groundedness Oracles
- Pre-Indexing Document ACL / RBAC Filtering
- Hallucination Span Detection (LettuceDetect)
- ScreenSpot & OSWorld GUI Grounding Evals
- NDCG@k · MRR · Recall@k Retrieval Audits
- Citation Span-Locking & Provenance Check
- MMMU · MathVista · DocVQA · OCRBench
- C2PA Cryptographic Media Watermarking

- FRAMES & LongBench Multi-Hop RAG Eval
- PII & Secret Redaction Before Indexing
- Visual Prompt-Injection OCR Scrubbing
- Bounding-Box IoU Click Accuracy Audit
- RGB (Retrieval-Augmented Generation Bench)
- Embedding Drift Monitor on Model Upgrade
- Audio Deepfake & Voice-Cloning Spoof Check
- Video Long-Horizon Needle-in-a-Haystack
- Cross-Tenant Vector Namespace Isolation
- Stale-Document TTL & Incremental Re-Index
- Voice Turn-Taking Latency (<300ms EoT)
- SynthID & AudioSeal Provenance Verification

Indirect Prompt Injection via RAG / Images
Naive Fixed-Size Mid-Table/Code Chunking
Visual OCR Hallucination & GUI Misclicks

▸ Poisoned Web/PDF Hijacking Agent via RAG
▸ Hidden White-on-White Text in Images
▸ 512-Token Split Severing Table Headers
▸ Unfiltered Vector Search Leaking Exec Docs

- Hallucinated Citations on Unretrieved Text
- VLM Misreading Dense Charts / Small Fonts
- Embedding Space Mismatch After Re-Index
- Coordinate Drift Clicking Wrong GUI Button